



AI chat assistant for law domain

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Abstract

Accessing and interpreting legal texts can be daunting for non-experts due to complex terminology and the sheer volume of legislation. While general-purpose Large Language Models (LLMs) offer conversational capabilities, they frequently generate plausible but factually incorrect legal information (hallucinations) when queried about specific national laws. In this project, we develop an AI chat assistant specialized in the Slovenian employment law domain aimed at providing comprehensible, strictly grounded answers. We utilized the COLESLAW 1.0 corpus, applying domain-specific filtering to construct a focused knowledge base. We designed a Retrieval-Augmented Generation (RAG) pipeline utilizing a BM25 lexical retriever combined with the Llama 3 model.

Keywords

Law domain, AI chatbot, LLMD4LH,

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Introduction

This project focuses on building an AI chat assistant for the Slovenian law domain, with a specific focus on employment law. The main goal is to help users get quick and understandable answers to legal questions while still keeping the answers grounded in official legal sources. We decided to narrow our scope to employment law because it is relevant and interesting topic for us. A narrower scope reduces the ambiguity often present in broader legal frameworks, minimizes the search space for the retrieval system, and makes qualitative evaluation more manageable and precise.

Existing Work

In recent years, several solutions have emerged that aim to simplify access to legal information and provide assistance in understanding national legislation. In the context of Slovenian law, we identified a small number of relevant platforms that partially address this problem.

One notable example is *Pravko.si*, which functions as a conversational assistant for legal questions. [1] The system provides relatively precise answers and, importantly, supports its responses with references to relevant legal acts and related materials. This increases transparency and allows users to verify the correctness of the information. Additionally, the assistant is capable of simplifying legal language to some extent, making it more accessible to non-expert users. However, we observed that the system struggles in cases where

the requested information is not available in its knowledge base, leading to unresponsive behavior instead of providing a fallback explanation or clarification.

Another platform is *MojMaliPravnik.net*, which operates more as a question-and-answer portal than an autonomous AI assistant. [2] Users can submit legal questions, and answers are either provided automatically or by human experts. Free questions and answers are published publicly (in anonymized form), creating a shared knowledge base that can benefit other users with similar issues. The automatically generated responses often resemble excerpts from legal documents, which may limit their readability and practical usefulness. For more complex inquiries, the platform offers paid individual legal advice, which is handled by professionals rather than automated systems.

From this overview, we observe that existing solutions either focus on conversational interaction with limited robustness, or rely heavily on human expertise with minimal automation. Furthermore, there is a lack of systems that combine accurate referencing of formal legal acts with reliable handling of missing information and consistently understandable explanations.

We also observe that general-purpose language models are ill-suited for Slovenian legal question answering. Trained on broad multilingual data, they lack specialized knowledge of Slovenian statutes and often hallucinate plausible but false information, such as inventing statute numbers. In practice, models like ChatGPT have been found to generate completely

fabricated case citations and legal quotes.

This gap motivates the development of our proposed AI assistant, which aims to provide grounded, explainable, and robust answers based directly on formal Slovenian legal sources while maintaining accessibility for general users.

Corpus Analysis

For the development of our system, we selected *COLESLAW 1.0*, a recent collection of Slovenian legal texts published in the CLARIN.SI repository. [3] The corpus contains 547,799 texts and 771,931,725 words distributed across 12 JSONL files. At the highest level, it combines four complementary legal domains: the PISRS legal information system, the *SodnaPraksa* collection of court decisions, the Constitutional Court collection (*USRS*), and texts from the *Uradni list*. This is useful for our task because the dataset does not focus only on laws or only on court decisions, but instead combines several types of legal texts in one place.

A closer look at the internal structure shows that the corpus is not balanced across domains, which is important for downstream modeling choices. The largest share by number of documents comes from *Uradni list*, which contributes 274,531 texts, or just over half of the corpus, while *SodnaPraksa* contributes 217,409 texts. In terms of word count, however, *SodnaPraksa* becomes the dominant source with 352.3 million words (45.6% of all words), followed by *Uradni list* with 296.7 million words (38.4%). PISRS is much smaller in the number of documents, with 40,749 texts, but still contributes 96.8 million words, which indicates that its documents are on average longer and denser than many gazette entries. The USRS collection adds a further 15,110 constitutional court documents and 26.2 million words.

The corpus also contains noticeable differences between subcollections. For example, *sp_courts* is the largest individual part of the dataset, while *ul-razglasni* contains shorter announcement-style texts. Some legislative collections in PISRS are much longer, especially the register of regulations and draft regulations. Because of this, the corpus includes both short notices and long legal documents. In our opinion, this makes COLESLAW 1.0 a good starting point for building a Slovenian legal assistant, since it is large, diverse, and based on official public sources.

To create a focused and highly relevant knowledge base, we filtered the comprehensive COLESLAW corpus to extract documents strictly pertaining to employment law. This was achieved by applying a set of domain-specific keywords (such as *delovno razmerje*, *odpovedni rok*, *letni dopust*, *minimalna plača*, etc.). This filtering process significantly reduced general legal noise and provided a clean, domain-specific subset of raw texts to serve as the foundation for our retrieval system.

System Implementation and Evaluation

To achieve our goals, we implemented a complete Retrieval-Augmented Generation (RAG) pipeline. The system consists

of two main components: a retriever and a generator.

For the retriever, we preprocess the filtered employment law documents by splitting them into fixed-size chunks (300 words). During this parsing phase, we also extract and preserve key metadata for each document—such as the name of the law, the specific article number, and the origin source. This metadata is strictly bound to its corresponding text chunk. We then build a search index using the BM25 Okapi algorithm. When a user asks a legal question, the retriever queries the index and fetches both the top relevant legal passages and their associated metadata.

For the generator, we provide the retrieved contextual passages along with their metadata to a Large Language Model (Llama 3) using a strict prompting strategy. This instructs the model to construct its response solely based on the retrieved texts and to explicitly cite the specific laws and articles provided in the metadata, ensuring all claims are traceable and transparent.

To measure the improvements brought by the RAG architecture, we established an automated evaluation pipeline. We curated a specialized dataset of 20 legal questions divided into factual, ambiguous, and unanswerable categories. Our evaluation methodology compares the answers generated by a baseline model (which relies only on its internal parameter knowledge) against our RAG-assisted system.

We employ an “LLM-as-a-judge” approach to systematically score the responses across several critical dimensions:

- **Factual correctness and grounding:** Measuring how accurately the answers reflect the provided legal acts.
- **Hallucination rate:** Penalizing the model if it fabricates legal claims or non-existent article numbers.
- **Refusal accuracy:** Testing the model’s ability to safely decline to answer when the provided legal documents do not contain the necessary information (unanswerable queries).

Additionally, we evaluate the retrieval quality independently (measuring the hit rate for factual questions), ensuring that any failures in the system can be correctly attributed to either the search index or the generative model.

RAG Example

To concretely illustrate our current RAG pipeline, consider the workflow when a user inputs the query: “*Koliko dni letnega dopusta mi pripada?*” (How many days of annual leave am I entitled to?).

First, the system queries the BM25 retrieval index, returning the top three most relevant text chunks from our filtered corpus. For instance, the top retrieved chunk correctly contains:

“*Delavec ima pravico do letnega dopusta, ki ne sme biti krajši od štirih tednov. Minimalni letni dopust za delavce, ki delajo pet dni na teden, znaša 20 delovnih dni...*” (ZDR-1, Article 159).

These retrieved chunks are then dynamically injected into a predefined prompt template as "context". The complete prompt is subsequently sent to the Llama 3 model via an API call (using the local Ollama or remote Open WebUI deployment). The prompt structure strictly guides the model:

```
System: You are an expert Slovenian
legal assistant. Answer the user's
question based strictly on the provided
Context. If the context does not
contain the answer, say so.
Context: [Injected top-3 retrieved
chunks with metadata]
User: Koliko dni letnega dopusta
mi pripada?
```

The model processes this combined input and generates a final response, successfully synthesizing the raw legal text into a user-friendly answer (explaining the 20-day minimum based on the five-day workweek) while properly citing ZDR-1, Article 159.

Evaluation Results

We executed our evaluation pipeline using the `llama3:latest` model via Ollama. The test targeted 20 questions, which were split into answerable factual queries and deliberately unanswerable queries designed to provoke failure. We evaluated both the "Baseline" (the model relying entirely on its intrinsic pre-trained knowledge) and the "RAG" setup.

Table 1 outlines the core performance metrics derived from our testing phase. The scoring methodology utilizes a 0 to 5 scale, where higher values indicate better performance for correctness, grounding, completeness, and clarity.

Table 1. Evaluation results summary comparing baseline generation against RAG.

Metric	llama3 baseline	llama3 RAG
Correctness	4.15	3.85
Grounding	4.75	4.95
Completeness	4.70	4.55
Clarity	4.95	4.85
Hallucination	1.00	0.25
Refusal Accuracy	0.00	0.75

The most compelling finding of our implementation is the mitigation of legal hallucinations. The baseline model registered a hallucination score of 1.00, demonstrating a propensity to confidently fabricate legal information when uncertain. Imposing RAG dynamics violently curtailed this behavior, driving the hallucination score down to just 0.25 (as seen in Figure 1).

Furthermore, the baseline model proved entirely incapable of admitting missing knowledge, achieving a 0.00 Refusal Accuracy. It forcefully attempted to answer every unanswerable

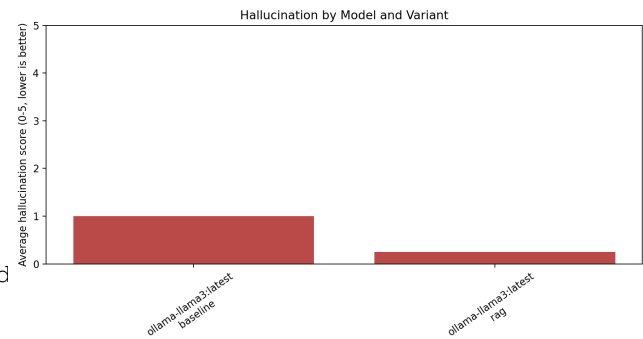


Figure 1. Hallucination rate comparison. The bar chart illustrates the stark difference in hallucination tendencies; the RAG architecture significantly minimizes fabricated legal claims compared to the baseline model (lower is better).

question. In stark contrast, the RAG system successfully exhibited a 0.75 Refusal Accuracy, accurately recognizing when the BM25 retrieval omitted relevant texts and safely declining to answer rather than providing dangerous misinformation.

Grounding marginally improved from 4.75 to 4.95 indicating the RAG model successfully utilized the parsed document metadata for formal citation. A slight decline in raw correctness and completeness scores (3.85 from 4.15) constitutes an expected trade-off: the RAG model acts significantly more conservatively, bound only to what it is provided, while the baseline model freely guesses broad (but unverified) concepts.

Retrieval Integrity

Our retrieval module underwent isolated testing entirely independently of the LLM generation phase. The BM25 Okapi index achieved an **87.5% Hit Rate**, effectively recovering relevant laws for the vast majority of our queries at an average context footprint of just 176.3 words. However, it generated a **25.0% False Evidence Rate** on unanswerable questions, meaning the system occasionally fetched completely irrelevant statutes simply because they shared lexical token overlap with the query. This indicates our architecture functions correctly at large but still demands improvement.

Analysis

To better understand the numerical metrics, a qualitative review of the generated responses reveals clear behavioral patterns. The RAG approach excels in scenarios requiring strict adherence to statutory limits and definitions. For example, when queried about statutory severance pay ("*Koliko znaša odpravnina?*"), the RAG system accurately grounds its answer in Article 108 of ZDR-1, explaining the exact fractional logic (e.g., 1/5 of the base salary per year of employment).

Conversely, the baseline model frequently falls victim to plausible-sounding legal hallucinations. When queried about specific rights or edge cases, the baseline model might authoritatively fabricate a rule—such as confidently claiming "the employer must provide exactly 3 days of extra leave for moving" or misstating the mandatory notice period—and back

it up by citing a completely non-existent or irrelevant article number from the ZDR-1. Our RAG implementation elegantly eliminates this behavior by forcing the model to either reliably cite the provided texts or admit ignorance.

Where the RAG system primarily fails is during “lexical mismatch” scenarios. If a user poses a question using colloquial synonyms (e.g., asking about “*bolniška*” instead of the formal statutory term “*začasna nezmožnost za delo*”), the exact-match BM25 search might fail to retrieve the relevant passage.

Consequently, the LLM receives empty or irrelevant context and defaults to its prompt constraint, refusing to answer. While this correctly avoids a hallucination, the resulting formal refusal is ultimately unhelpful to the user, portraying a brittle retrieval step.

Future Improvements

While the current methodology establishes a robust baseline for legal question answering, we identify several areas for future enhancements. At present, the retrieval component relies exclusively on exact lexical matching, meaning that legal documents can naturally be overlooked if the user phrases their query using synonyms or colloquial terms that differ from the strict formal language of the statutes.

To address this limitation, we plan to explore the integration of dense vector embeddings and semantic search. Incorporating models capable of deeply understanding the semantic meaning behind questions would allow the system to fetch relevant documents even when there are no exact keyword overlaps, significantly increasing the recall and overall robustness of the retriever. Furthermore, employing a hybrid retrieval approach—combining the precision of BM25 with the semantic recall of vector embeddings—paired with a cross-encoder reranker, represents the most promising direction for elevating the assistant’s factual accuracy in our next development phases.

References

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